use the LASER embeddings (Schwenk and Douze, 2017), which provide language-agnostic sentence representations, in order to assess cross-lingual similarity. We mine n-gram pairs from the Europarl corpus, which consists of around 20 million parallel sentences extracted from the proceedings of the European Parliament (Koehn, 2005). For all other tasks, we use E5 (Wang et al., 2022), which are multilingual sentence representations trained using contrastive learning on a diverse range of tasks. For TriviaQA, we mine the n-gram pairs from TriviaQA's training set. For MMLU, since the training set is very small ($100 - 500$ examples for each task), we mine the n-gram pairs from the test sets directly. For GSM8K, due to the short time limitation during rebuttal, we also mine the n-gram pairs from the 1K test sets directly.

In WMT, the cosine similarity thresholds we use are: 0.85, 0.8, 0.75, and 0.7 for 2 to 5-gram pairs, respectively; For TriviaQA and MMLU, the values are 0.75 and 0.65 for 3 and 5-gram pairs, respectively. We use lower thresholds for larger n-grams because larger n-grams inherently impose stricter alignment, and are therefore less likely. We show a cosine similarity sensitivity ablation in Figure 6.

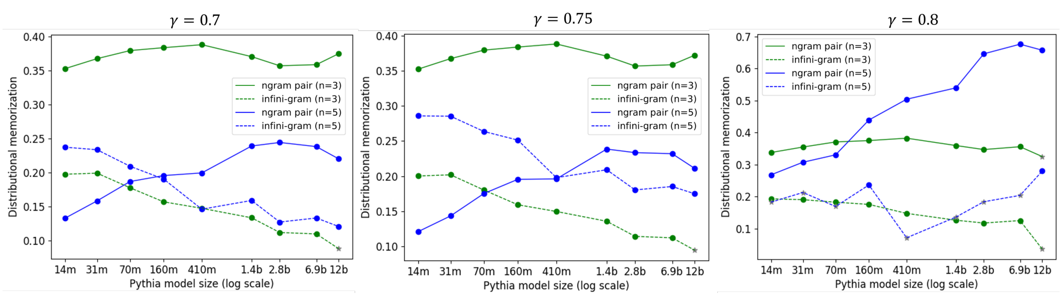

Figure 6: Visualization of distributional memorization with different-sized Pythia models on TriviaQA with different cosine similarity threshold $\gamma \in \{0.7, 0.75, 0.8\}$. The trend of distributional memorization does not change with different thresholds.

We perform our experiments on 8 GPU 40G A100 working stations. Below is the license information for the datasets we used:

- Pile: MIT license. URL: `https://github.com/EleutherAI/the-pile/tree/master`
- Tulu: ODC-BY license. URL: `https://huggingface.co/datasets/allenai/tulu-v2-sft-mixture`
- WMT-09: published with the WMT workshop. URL: `https://www.statmt.org/wmt09/translation-task.html`
- TriviaQA: Apache License 2.0. URL: `https://nlp.cs.washington.edu/triviaqa/`
- MMLU: MIT license. URL: `https://github.com/hendrycks/test`

Knowledge-intensive MMLU tasks:

```
'prehistory', 'business_ethics', 'philosophy',
'moral_disputes', 'medical_genetics', 'high_school_government_and_politics',
'human_aging', 'us_foreign_policy', 'high_school_macroeconomics',
'logical_fallacies', 'international_law', 'computer_security',
'sociology', 'professional_psychology', 'marketing', 'human_sexuality',
'anatomy', 'high_school_us_history', 'public_relations',
'high_school_microeconomics', 'clinical_knowledge', 'security_studies',
'nutrition', 'world_religions', 'high_school_psychology',
'high_school_geography', 'management', 'global_facts',
'high_school_world_history', 'high_school_european_history',
'jurisprudence', 'virology', 'astronomy', 'miscellaneous'
```

Reasoning-intensive MMLU tasks:

'econometrics', 'professional_law', 'abstract_algebra', 'college_medicine', 'college_chemistry', 'moral_scenarios', 'college_mathematics', 'high_school_chemistry', 'professional_accounting', 'college_computer_science', 'college_biology', 'high_school_computer_science', 'high_school_mathematics', 'college_physics', 'professional_medicine', 'elementary_mathematics', 'machine_learning', 'electrical_engineering', 'high_school_physics', 'conceptual_physics', 'high_school_statistics', 'high_school_biology'

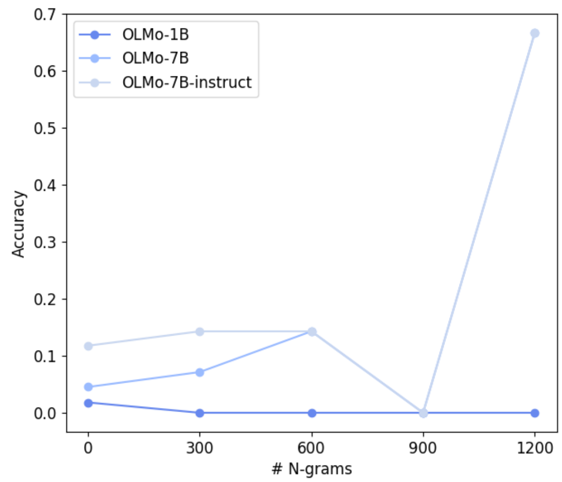

Figure 7: GSM8K accuracy v.s. n-gram pair count in Dolma with different OLMo model sizes.

E Prompt Optimization Details

Corpus	Dataset	Memorization	Generalization
Pile	**TriviaQA**	Deliver an exact single word or concise phrase in response to the factual question. (avg. 3-gram count: 557948.5)	Formulate a distinctive and concise term or phrase to clearly answer the factual question. (avg. 3-gram count: 1714.9)
	GSM8K	Carefully analyze each math word problem presented, break it down step-by-step, and clearly state the final answer. (avg. 3-gram count: 45028.9)	Dissect each math word problem into straightforward, logical steps; solve each part systematically for precise solutions. (avg. 3-gram count: 43.1)
Dolma	**TriviaQA**	Provide a single word or concise phrase in response to the following factual question. (Avg. 3-gram count: 5566475.5)	Provide a clear and specific word or brief phrase in response to the factual question below. (Avg. 3-gram count: 43436.2)
	GSM8K	Solve the following math word problem by methodically breaking it down into simple, clear steps to find the correct solution efficiently. (Avg. 3-gram count: 3736735.0)	Solve the upcoming math word problem by sequentially explaining each calculation and logical step, ensuring clarity and coherence in your solution. (Avg. 3-gram count: 4563.6)

Table 2: Prompts optimized for memorization and generalization for TriviaQA and GSM8K.

Text in the square brackets are comments that are not involved in the actual prompt.

Meta prompt for prompt optimization

```
**Task Description**:

You are tasked with optimizing a given prompt to guide an open-source language model (LM)
    in completing a specific task effectively. You will receive:

- The current prompt for the task.
- Its corresponding memorization score (Average frequency of task-related n-grams found in
    the LM's pretraining corpus).
- A few example input-output pairs illustrating the intended task.
- A history of previous prompt optimization iterations.

**Optimization Goals**:

Clearly describe the intended task with a general instruction that effectively guides the
    LM to perform the task.

[If trying to encourage reasoning ...]

Minimizing the memorization score of the updated prompt. The memorization score reflects
    the distributional correlation between the prompt and the LM's pretraining corpus. A
    lower score encourages the LM to generate more novel outputs.

[If trying to encourage memorization ...]

Maximize the memorization score of the updated prompt. The memorization score reflects the
    distributional correlation between the prompt and the LM's pretraining corpus. A
    higher score encourages better alignment with the LM's learned knowledge.

**Example task input-output pairs**:

[Example input-output pairs for current task]

**Output**:

Produce an updated prompt that balances clarity of task instruction with an lower
    memorization score.
```

F n-GRAM PAIR EXAMPLES

In this section, we present some representative examples collected from the analysis for the different tasks evaluating, including Translation and Question-Answering (MMLU, TriviaQA). In order to show examples from the different experiments, we show examples with different model sizes and number of n-grams.

G MODEL GENERATIONS VS. n-GRAM FREQUENCY IN PRETRAINING

To further analyze the behavior of a model based on the pairs found in pretraining, we compared the alignment of the generations of the model when that n-gram was generated, vs. n-gram frequency in the pretraining corpus. We found that as the n-gram frequency increased, the model n-gram pairs the model generated were more aligned.